Inference often combines conditioning and summarizing

Observed y $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$ $\rightarrow$ **Summarize** $\hat{z} = E[z \mid y] = \mu_{z|y}$

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z|y}$

Choose a reconstruction by its conditional squared-error risk

Squared-error risk

Observed y; $z, a \in \mathbb{R}^d$

$$L(a, z) = \|z - a\|^2$$

$$R(a|y) = E[L(a, z)|y]$$

Centered error

$$m = E[z|y], \quad r = z - m$$

$$z - a = r + (m - a)$$

Squared-error expansion

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, offset $m - a$ is fixed; uncertainty remains in r.

Centering separates residual uncertainty from estimation error

Conditional expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

Cross-term

$$E[r|y] = E[z|y] - m = 0$$
$$E[r^T(m-a)|y] = E[r|y]^T(m-a) = 0$$

Residual energy

$$V = \mathrm{Cov}[z|y]$$
$$E[\|r\|^2|y] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$

- Risk is residual uncertainty plus squared distance from the posterior mean.

The posterior mean minimizes squared-error risk

Conditional squared-error risk Previous slide: residual uncertainty plus squared distance from the mean

$$R(a|y) = \operatorname{tr} V + \|m - a\|^2$$

Minimizing action

$$\|m - a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y]$$

$$\min_a R(a|y) = \operatorname{tr} V$$

Conditional Gaussian

$$a^*(y) = \mu_{z|y}$$

$$\min_a R(a|y) = \operatorname{tr} \Sigma_{z|y}$$

- The conditional mean minimizes model expected squared error.
- The posterior covariance describes uncertainty that remains after the mean action is chosen.

Choose a reconstruction by its conditional squared-error risk

Observed y; reconstruct $z \in \mathbb{R}^d$ with $a \in \mathbb{R}^d$

Squared-error criterion

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center: $m = E[z|y]$, $r = z - m$

$$z - a = r + (m - a)$$

Square the centered error

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, offset $m - a$ is fixed; uncertainty remains in r.

Centering separates residual uncertainty from estimation error

Conditional expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

Cross-term

$$E[r|y] = E[z|y] - m = 0$$
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Residual energy

$$V = \mathrm{Cov}[z|y]$$
$$E[\|r\|^2|y] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$

- Risk is residual uncertainty plus squared distance from the posterior mean.

The posterior mean minimizes squared-error risk

$$R(a|y) = \underbrace{\operatorname{tr} V}_{\text{independent of a}} + \underbrace{\|m-a\|^2}_{\text{nonnegative; zero iff } a=m}$$

Consequences at the minimizing action

Minimum risk

$$\min_a R(a|y) = \operatorname{tr} V$$

Optimal action

$$a^*(y) = m = E[z|y]$$

For the conditional Gaussian

$$\min_a R(a|y) = \operatorname{tr} \Sigma_{z|y} \qquad a^*(y) = \mu_{z|y}$$

- The conditional mean minimizes model expected squared error.
- The posterior covariance describes uncertainty that remains after the mean action is chosen.